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Fed-ADE: Adaptive Learning Rate for Federated Post-adaptation under Distribution Shift

Heewon Park[1]*, Mugon Joe[2]*, Miru Kim[1]*, Kyungjin Im[2], Minhae Kwon[1]†
[1] Department of Electrical and Computer Engineering, Sungkyunkwan University, Republic of Korea
[2] Soongsil University, Seoul, Republic of Korea

Abstract

*Federated learning (FL) in post-deployment settings must adapt to non-stationary data streams across heterogeneous clients without access to ground-truth labels. A major challenge is learning rate selection under client-specific, time-varying distribution shifts, where fixed learning rates often lead to underfitting or divergence. We propose Fed-ADE (**Fed**erated **A**daptation with **D**istribution Shift **E**stimation), an unsupervised federated adaptation framework that leverages lightweight estimators of distribution dynamics. Specifically, Fed-ADE employs uncertainty dynamics estimation to capture changes in predictive uncertainty and representation dynamics estimation to detect covariate-level feature drift, combining them into a per-client, per-timestep adaptive learning rate. We provide theoretical analyses showing that our dynamics estimation approximates the underlying distribution shift and yields dynamic regret and convergence guarantees. Experiments on image and text benchmarks under diverse distribution shifts (label and covariate) demonstrate consistent improvements over strong baselines. These results highlight that distribution shift-aware adaptation enables effective and robust federated post-adaptation under real-world non-stationarity. The code is available at* https://github.com/h2w1/Fed-ADE

1. Introduction

Machine learning models are increasingly deployed across edge devices such as smartphones, IoT sensors, and autonomous systems, where each device continuously receives a non-stationary data stream [25, 46, 59, 60]. FL has emerged as a key paradigm to support such decentralized learning under privacy constraints, enabling model collaboration without raw data sharing [12, 21, 30, 31, 35, 36, 44, 51, 61]. In practice, models are pre-trained on centralized datasets and deployed to client devices, but their performance often degrades quickly due to distribution shifts in the real-time data. These shifts appear in multiple forms, such as label distribution shift and covariate shift, and evolve in client-specific and heterogeneous ways [16, 20, 27, 28, 33, 43, 49].

This environment introduces two major sources of heterogeneity: (i) shift heterogeneity, where each client experiences different temporal dynamics of distribution shift [8, 20, 22, 27, 41, 43, 68], and (ii) data heterogeneity, where clients' local datasets already differ in scale, domain, or semantics [12, 36, 47, 51, 61]. Together, these factors make federated post-deployment adaptation highly challenging, as a single global model or fixed training learning rate is insufficient to cope with diverse and evolving conditions.

A critical yet underexplored aspect of this problem is the choice of learning rate. In online adaptation, learning rate selection directly governs both stability and responsiveness. Fixed learning rates fail to reflect data distribution shifts, leading to underfitting when the learning rate is too small, or divergence when the learning rate is too large [2, 3, 11, 42]. The difficulty is amplified in FL, where hundreds of heterogeneous clients undergo distinct, time-varying distribution shifts without access to ground-truth labels.

To address this, we propose Fed-ADE, a lightweight and unsupervised adaptation framework. Figure 1 shows the overview of the proposed Fed-ADE system. Fed-ADE introduces two estimators of distribution dynamics: *Uncertainty dynamics estimation*, which captures predictive uncertainty changes, and *representation dynamics estimation*, which detects embedding-level drift. By combining these signals, Fed-ADE assigns a per-client, per-timestep adaptive learning rate that reflects local distribution. Unlike prior approaches relying on multi-model ensembles or costly hyperparameter search, Fed-ADE requires no extra communication or supervision. Furthermore, we provide theoretical analyses showing that dynamics estimation approximates underlying distribution evolution and yields dynamic regret guarantees. Empirically, we validate Fed-ADE across image and text benchmarks under label and covariate shift environments, where it consistently improves adaptation accuracy and efficiency compared to strong baselines.

*Equal contribution.
†Corresponding author. minhae.kwon@skku.edu

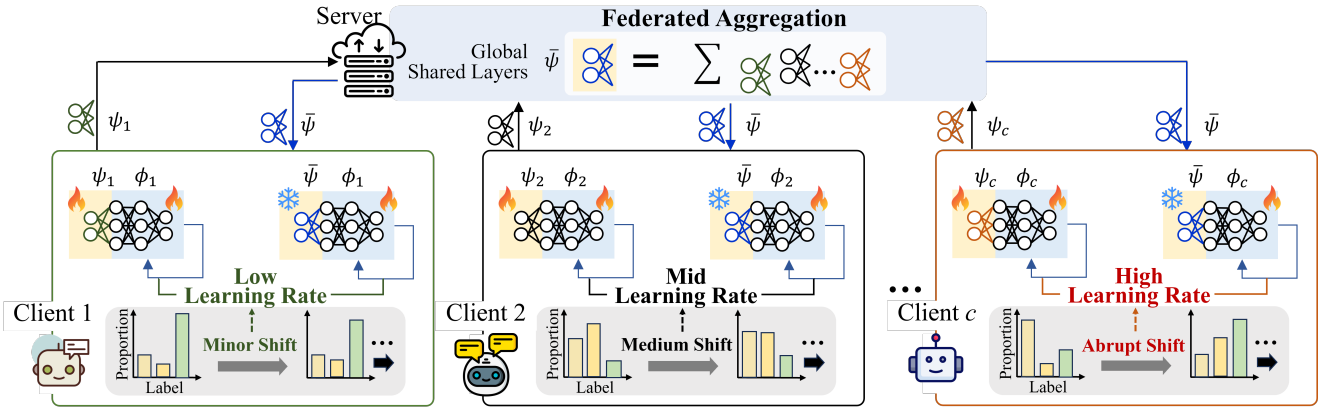


Figure 1. Overview of the Fed-ADE. Each client receives a pre-trained model, splits it into shared (ψ_c) and personalized (ϕ_c) layers, and adapts to unlabeled, distribution-shifting data using an adaptive learning rate.

In summary, our contributions are as follows.

- We propose Fed-ADE, a novel framework for unsupervised federated post-adaptation under heterogeneous and time-varying distribution shifts, addressing real-time data distribution shifts.
- We introduce two lightweight estimators: Uncertainty dynamics estimation, which captures predictive uncertainty changes, and representation dynamics estimation, which detects embedding-level drift. These estimators enable per-client, per-timestep adaptive learning rates without requiring ground-truth labels.
- We provide theoretical analyses showing that distribution shift estimation approximates true distribution shift and that the resulting learning dynamics satisfy dynamic regret bound and convergence guarantees.
- We conduct extensive experiments on four image and a text benchmarks under label and covariate shift, demonstrating that Fed-ADE consistently outperforms state-of-the-art baselines while remaining computationally efficient.

2. Related Works

Federated Learning for Decentralized Data Environments FL is a widely adopted framework for collaboratively training machine learning models across multiple decentralized clients while preserving data privacy [12, 23, 34, 35, 44, 61]. Early approaches such as FedAvg [44] proposed aggregating locally updated model parameters on a central server without sharing raw data. However, this paradigm encounters challenges caused by data heterogeneity, as each client may hold data drawn from distinct distributions [23, 34, 35]. To mitigate these issues, personalized FL methods have been introduced to better adapt models to the local characteristics of each client's data. Among them, partial-sharing strategies train only a subset of layers collaboratively via FL, while the remaining layers are updated locally [12, 27, 51, 61]. These methods enhance personalization and local adaptation, yet they commonly assume stationary or globally synchronized data distributions across clients. In contrast, our work introduces a novel system that supports post-deployment adaptation by employing an unsupervised, per-client adaptive learning rate mechanism, which dynamically adjusts to real-time distributional shifts.

Distribution Shift Adaptation Distribution shift arises in various forms, including label and covariate shift [13, 15, 26, 53, 55, 64]. Online label shift adaptation adjusts predictions without true labels, as in ATLAS [2] and FTH [58]. Covariate shift is commonly addressed by domain adaptation methods such as UDA [18] and UNIDA [63], which mainly assume centralized settings and are not readily applicable to federated or online scenarios. In federated environments, adaptation approaches such as Fed-POE [43], ATP [4], and FedSPL [38] mitigate client heterogeneity or adapt models to unlabeled target clients through ensemble strategies, adaptation-specific rules, or pseudo-labeling. However, these methods often introduce additional computational overhead or auxiliary components during adaptation.

3. Preliminaries

Pre-training ($t = 0$) Let θ be the global model trained at the server using pre-collected data $(\mathbf{x}_{\mathcal{G}}^0, \mathbf{y}_{\mathcal{G}}^0)$, where $\mathbf{x}_{\mathcal{G}}^0$ follows feature distribution $\mathbf{Q}_{\mathcal{G},\mathbf{x}}^0$ and $\mathbf{y}_{\mathcal{G}}^0$ follows label distribution $\mathbf{Q}_{\mathcal{G},\mathbf{y}}^0$. We also denote by $\mathbf{Q}_{\mathcal{G},\mathbf{x}|\mathbf{y}}^0$ the conditional distribution of features given labels, and by $\mathbf{Q}_{\mathcal{G},\mathbf{y}|\mathbf{x}}^0$ the conditional distribution of labels given features. Each label $\mathbf{y}_{\mathcal{G}}^0$ is an index i, where i is an element of the class index set $\mathcal{I}$. In the distribution $\mathbf{Q}_{\mathbf{y}_{\mathcal{G}}^0}$, the i-th component, $[\mathbf{Q}_{\mathbf{y}_{\mathcal{G}}^0}]_i$, represents the proportion of class $i \in \mathcal{I}$ in the label set $\mathbf{y}_{\mathcal{G}}^0$.

Online Distribution Shift ($0 < t \leq T$) Consider a set of federated clients $\mathcal{C}$ collaboratively performing post-deployment adaptation. Each client model θ_c is initialized from the pre-trained global model θ. At time t, client $c \in \mathcal{C}$ observes an unlabeled data stream $\mathbf{x}_c^t$ drawn from a client and

time-dependent distribution $\mathbf{Q}^t_{c,\mathbf{x}}$, while the corresponding labels $\mathbf{y}^t_c$ follow $\mathbf{Q}^t_{c,\mathbf{y}}$ but remain unobserved. For brevity, we use $\mathbf{Q}^t_c$ to denote the overall data distribution at time t.

A temporal distribution shift occurs when $\mathbf{Q}^t_c \neq \mathbf{Q}^{t-1}_c$. Typical cases include: (1) label distribution shift, where $\mathbf{Q}^t_{c,\mathbf{y}}$ changes while $\mathbf{Q}^t_{c,\mathbf{x}|\mathbf{y}}$ stays approximately stable and (2) covariate shift, where $\mathbf{Q}^t_{c,\mathbf{x}|\mathbf{y}}$ changes while $\mathbf{Q}^t_{c,\mathbf{y}}$ is approximately constant. Following prior works [2, 58, 62], we model the shift of $\mathbf{Q}^t_c$ as

$$\mathbf{Q}^t_c = (1-\omega(t))\,\mathbf{Q}^0_c + \omega(t)\,\mathbf{Q}^T_c, \tag{1}$$

where $\omega(t) \in [0,1]$ is a weighting function interpolating between the initial distribution $\mathbf{Q}^0_c$ and a later distribution $\mathbf{Q}^T_c$. In the federated setting, we assume that $\mathbf{Q}^0_c$ aligns with the pre-training data distribution, whereas $\mathbf{Q}^T_c$ represents each client's distinct distribution evolved after deployment, reflecting client-specific environments and behaviors.

In this setting, an unsupervised federated post-adaptation framework is required to adjust to the continuously changing data distributions with limited client samples, while maintaining robust global performance.

Learning Objective with Unsupervised Risk Estimation Under the federated setting with online distribution shift, each client $c \in \mathcal{C}$ aims to update its local model θ_c using only unlabeled data. We adopt an unsupervised risk estimation framework that enables model updates without access to ground-truth labels. Let the expected risk for client c at time t be defined as

$$\begin{aligned}\mathcal{F}^t_c(\theta_c) &\triangleq \mathbb{E}_{\mathbf{y}^t_c \sim \mathbf{Q}^t_{c,\mathbf{y}}}\left[\mathcal{L}\Big(\mathcal{H}(\theta_c;\mathbf{x}^t_c), \mathbf{y}^t_c\Big)\right] \\ &= \sum_{i\in\mathcal{I}} [\mathbf{Q}^t_{c,\mathbf{y}}]_i \cdot \mathcal{F}^{t,i}_c(\theta_c),\end{aligned} \tag{2}$$

where $\mathcal{H}(\cdot)$ denotes the softmax prediction of the model, $\mathcal{L}(\cdot)$ is the loss function, and $\mathcal{F}^{t,i}_c(\theta_c)$ is the class-wise risk for class i, which can be expressed as $\mathbb{E}_{\mathbf{y}^t_c \sim [\mathbf{Q}^t_{c,\mathbf{y}}]_i}\big[\mathcal{L}\big(\mathcal{H}(\theta_c;\mathbf{x}^t_c), \mathbf{y}^t_c\big)\big]$. The expected risk can be decomposed as

$$\mathcal{F}^t_c(\theta_c) = \sum_{i\in\mathcal{I}} [\mathbf{Q}^t_{c,\mathbf{y}}]_i\, \mathcal{F}^{0,i}_c(\theta_c), \tag{3}$$

where $\mathcal{F}^{0,i}_c(\theta_c)$ denotes the class-wise risk computed from the initial data $(\mathbf{x}^0_c, \mathbf{y}^0_c)$, which is assumed to follow the same distribution as the global pre-training data. Hence, $\mathcal{F}^{0,i}_c(\theta_c)$ can be approximated by its empirical counterpart $\widehat{\mathcal{F}}^{0,i}_c(\theta_c)$ using $(\mathbf{x}^0_c, \mathbf{y}^0_c)$.

Since labels are not observable during post-adaptation, the current label distribution $\mathbf{Q}^t_{c,\mathbf{y}}$ must be estimated. We employ the Black-box Shift Estimation (BBSE) method [40] for this purpose, which corrects the pseudo-label distribution via $\mathbf{Q}^t_{c,\mathbf{y}} \approx \mathbf{M}^{-1}\mathbf{Q}^t_{c,\hat{\mathbf{y}}}$. Here, $\mathbf{M}$ is a confusion matrix computed on the server using pre-training data $(\mathbf{x}^0_{\mathcal{G}}, \mathbf{y}^0_{\mathcal{G}})$ and shared with clients, while $\mathbf{Q}^t_{c,\hat{\mathbf{y}}}$ denotes the prediction distribution over the unlabeled batch $\mathbf{x}^t_c$.[1] Using these estimates, the unsupervised risk can be approximated as

$$\widehat{\mathcal{F}}^t_c(\theta_c) = \sum_{i\in\mathcal{I}} \left[\widehat{\mathbf{M}}^{-1}\mathbf{Q}^t_{c,\hat{\mathbf{y}}}\right]_i \cdot \widehat{\mathcal{F}}^{0,i}_c(\theta_c), \tag{4}$$

where $\widehat{\mathbf{M}}$ is empirical estimator of the confusion matrix.

Finally, the federated learning objective based on the estimated unsupervised risk in (4) is to jointly optimize all client models as follows.

$$\{\theta^*_c\}_{c=1}^C = \arg\min_{\theta_c} \sum_{c=1}^C \widehat{\mathcal{F}}^t_c(\theta_c) \tag{5}$$

Personalized Federated Learning with Layer Decoupling To achieve the goal in (5) under heterogeneous client distributions (i.e., $\mathbf{Q}^t_c$ may differ across c and shift over time), we adopt a partial-sharing FL method that decouples each client model θ_c into shared layers ψ_c and personalized layers ϕ_c, where only the shared layers ψ_c are communicated with the server while the personalized layers ϕ_c are retained locally [12, 27, 39, 47, 51, 61]. Specifically, we adopt two step local update method [12, 27, 51, 61]. Each client participates in R rounds of federated communication. In round $r \in \{1,\dots,R\}$, a subset of clients $\mathcal{C}^{(r)} \subset \mathcal{C}$ updates both shared and personalized layers locally with the learning rate η^t_c, and the risk estimator $\widehat{\mathcal{F}}^t_c\big(\{\psi_c,\phi_c\}\big)$ obtained in (4) as follows.

$$\{\psi_c,\phi_c\} \leftarrow \{\psi_c,\phi_c\} - \eta^t_c \nabla_{\psi_c,\phi_c} \widehat{\mathcal{F}}^t_c\big(\{\psi_c,\phi_c\}\big) \tag{6}$$

Then, each clients sends ψ_c to the server, which aggregates them using a weighted average based on each client's local dataset size N^t_c as follows.

$$\bar{\psi} = \frac{1}{\sum_{c\in\mathcal{C}^{(r)}} N^t_c} \sum_{c\in\mathcal{C}^{(r)}} N^t_c\, \psi_c \tag{7}$$

The server broadcasts $\bar{\psi}$ to all clients, who set $\psi_c \leftarrow \bar{\psi}$ and thus hold $\theta_c = \{\bar{\psi},\phi_c\}$. After receiving $\bar{\psi}$, clients freeze the shared layers and adapt only the personalized layers as follows.

$$\phi_c \leftarrow \phi_c - \eta^t_c \nabla_{\phi_c} \widehat{\mathcal{F}}^t_c\big(\{\psi_c,\phi_c\}\big) \tag{8}$$

With this update strategy, clients can preserve aggregated global knowledge in $\bar{\psi}$ while capturing the client-specific structure in ϕ_c.

[1] Details of BBSE and the derivation of the risk estimator are provided in Appendix B.

Algorithm 1 Fed-ADE

Inputs: Total rounds R, and learning rate bounds $[\eta_{\min}, \eta_{\max}]$
Initialize: Local model $\{\psi_c, \phi_c\}$, previous summaries $(\mathbf{q}_c^{t-1}, \mathbf{z}_c^{t-1})$
for $r = 1, 2, \ldots, R$ **do**
  Server samples participants $\mathcal{C}^{(r)} \subset \mathcal{C}$
  for $c \in \mathcal{C}^{(r)}$ **in parallel do**
    Build unsupervised risk $\widehat{\mathcal{F}}_c^t(\psi_c, \phi_c)$ as (4)
    Compute batch summaries $\mathbf{q}_c^t$ and $\mathbf{z}_c^t$
    Estimate dynamics $\mathcal{S}_{\text{unc},c}^t$ as (12), $\mathcal{S}_{\text{rep},c}^t$ as (14)
    Combine shift dynamic signals $\mathcal{S}_c^t$ as (10)
    Set adaptive learning rate η_c^t as (9)
    Update local model $\{\psi_c, \phi_c\}$ by (6)
    Send ψ_c to server
  end for
  Server aggregates to obtain $\bar{\psi}$ via (7)
  Server broadcasts $\bar{\psi}$
  for $c \in \mathcal{C}^{(r)}$ **in parallel do**
    Replace shared layers $\psi_c \leftarrow \bar{\psi}$
    Update personalized layers ϕ_c as (8)
    Cache $(\mathbf{q}_c^{t-1}, \mathbf{z}_c^{t-1}) \leftarrow (\mathbf{q}_c^t, \mathbf{z}_c^t)$
  end for
end for

4. Distribution Shift Driven Federated Post-adaptation

In this section, we introduce Fed-ADE, a federated post-adaptation framework designed to address heterogeneous and time-varying distribution shifts across clients. Fed-ADE adaptively controls learning rates of each client based on the degree of local distribution shift, enabling fast yet stable post-deployment adaptation. The overall process of Fed-ADE is shown in Algorithm 1.

4.1. Distribution Shift Adaptive Learning Rate

The effectiveness of adaptation in (6) and (8) hinges on selecting a learning rate η_c^t that reflects how rapidly the local data distribution shifts at client c at timestep t. To capture this temporal shift, Fed-ADE introduces a distribution dynamics signal $\mathcal{S}_c^t$ that quantifies the magnitude of the observed distribution shift. The learning rate is then adaptively scheduled according to this signal as follows.

$$\eta_c^t = \eta_{\min} + (\eta_{\max} - \eta_{\min})\, \mathcal{S}_c^t \tag{9}$$

The adaptive rate η_c^t is bounded within $[\eta_{\min}, \eta_{\max}]$. A higher $\mathcal{S}_c^t$ indicates stronger distribution shift, resulting in a larger η_c^t for faster adaptation, whereas a smaller $\mathcal{S}_c^t$ implies stability, leading to a lower η_c^t for conservative updates. This mechanism allows Fed-ADE to automatically balance adaptivity and stability in real-time across heterogeneous clients.

4.2. Distribution Shift Dynamics Signal

We next describe how the dynamics signal $\mathcal{S}_c^t$ is constructed. At each timestep t, client c estimates the extent of its local distribution shift using two complementary components: (i) an uncertainty-based estimator $\mathcal{S}_{\text{unc}}^t$ that captures predictive uncertainty changes, and (ii) a representation-based estimator $\mathcal{S}_{\text{rep}}^t$ that measures embedding-level drift. These signals are combined to form a unified signal as follows.

$$\mathcal{S}_c^t = \frac{1}{2}\left(\mathcal{S}_{\text{unc}}^t + \mathcal{S}_{\text{rep}}^t\right) \in [0, 1] \tag{10}$$

Here, larger $\mathcal{S}_c^t$ indicates stronger local drift, prompting a corresponding increase in the adaptive learning rate.

Uncertainty Dynamics Estimation To capture temporal changes in predictive uncertainty, each client summarizes its model outputs over the current data batch $\mathbf{x}_c^t$ using the mean softmax vector as

$$\mathbf{q}_c^t = \frac{1}{|\mathbf{x}_c^t|} \sum_{x \in \mathbf{x}_c^t} \mathcal{H}(\{\psi_c, \phi_c\}; x), \tag{11}$$

where $\mathbf{q}_c^t$ represents the aggregated predictive distribution. This serves as an entropy-based proxy for uncertainty, providing a label-free summary of class-level belief of the model. Batch-level averaging normalizes for batch size and mitigates per-sample stochasticity. The temporal change in predictive uncertainty is then measured via the cosine distance between consecutive batches as follows.

$$\mathcal{S}_{\text{unc}}^t = 1 - \cos\left(\mathbf{q}_c^{t-1}, \mathbf{q}_c^t\right) \in [0, 1] \tag{12}$$

This yields values in $[0, 1]$ since both $\mathbf{q}_c^{t-1}$ and $\mathbf{q}_c^t$ are normalized probability vectors. Smaller $\mathcal{S}_{\text{unc},c}^t$ indicates stable predictions with low uncertainty drift, whereas values closer to 1 reflect stronger predictive changes. Computation requires storing only $\mathbf{q}_c^{t-1}$, incurring $O(|\mathcal{I}|)$ memory overhead.

Representation Dynamics Estimation To capture shifts in the embedding space, Fed-ADE summarizes the feature representation as an ℓ_2-normalized batch mean, which ensures that cosine distance reflects only directional changes rather than scale differences, as

$$\mathbf{z}_c^t = \frac{1}{|\mathbf{x}_c^t|} \sum_{x \in \mathbf{x}_c^t} \frac{h_{\psi_c}(x)}{\|h_{\psi_c}(x)\|_2} \in \mathbb{R}^d, \tag{13}$$

where $h_{\psi_c}(\cdot)$ denotes the feature extractor parameterized by ψ_c, and $\mathbf{z}_c^t$ represents the batch-level latent feature vector.

The magnitude of the feature-level shift is then computed as follows.

$$\mathcal{S}^t_{\text{rep}} = \tfrac{1}{2}\big(1 - \cos\big(\mathbf{z}_c^{t-1}, \mathbf{z}_c^t\big)\big) \in [0,1] \tag{14}$$

The scaling value $\frac{1}{2}$ normalizes the cosine distance from its natural range $[0,2]$ to $[0,1]$, aligning it with the uncertainty dynamics signal $\mathcal{S}^t_{\text{unc}}$. This estimator is fully label-free, operates locally, and only requires storing $\mathbf{z}_c^{t-1}$, incurring $O(d)$ memory overhead.

5. Theoretical Analyses

We provide theoretical analyses underpinning Fed-ADE. We first establish that the cumulative uncertainty and representation dynamics surrogates based on cosine similarity accurately approximate their respective true distribution shifts in unlabeled federated settings. These cumulative quantities are then used to derive a dynamic regret bound, in which each client's adaptive learning rate η_c^t is determined by its instantaneous combined shift signal $\mathcal{S}_c^t$, achieving the standard min–max optimal rate under non-stationarity. The analysis that verifies the adaptive updates in Fed-ADE satisfy convergence guarantees under non-stationary environments is provided in Appendix F.

5.1. Cumulative Shift Surrogates and Error Bounds

Cumulative Uncertainty Dynamics The true cumulative deviation of the predictive-distribution marginal $\sum_{t=1}^T \|\mathbf{Q}^t_{c,\mathbf{y}} - \mathbf{Q}^{t-1}_{c,\mathbf{y}}\|_1$ is unobservable in practice. We instead use the surrogate $\bar{\mathcal{S}}_{\text{unc}} \triangleq \sum_{t=1}^T \mathcal{S}^t_{\text{unc}}$. Under standard calibration, concentration, and non-degeneracy assumptions, $\bar{\mathcal{S}}_{\text{unc}}$ tracks the true predictive-distribution path length up to a small additive error.

Theorem 1 (Error Bound: Cumulative Uncertainty Dynamics)**.** *Assume ϵ-calibration with respect to $\mathbf{Q}^t_{c,\mathbf{y}}$, i.e., $\|\mathcal{H}(\{\psi_c,\phi_c\};\mathbf{x}_c^t) - \mathbf{y}_c^t\|_2 \le \epsilon_t$ in expectation. Then, the cumulative estimation error is bounded as*

$$\Big|\bar{\mathcal{S}}_{\text{unc}} - \textstyle\sum_{t=1}^T \|\mathbf{Q}^t_{c,\mathbf{y}} - \mathbf{Q}^{t-1}_{c,\mathbf{y}}\|_1\Big| \;\le\; K_{\cos} \textstyle\sum_{t=2}^T (\epsilon_t + \epsilon_{t-1}), \tag{15}$$

where $K_{\cos}$ is the Lipschitz constant of the cosine similarity function.

Proof. See Appendix C. □

Cumulative Representation Dynamics For representation-level drift, we define $\bar{\mathcal{S}}_{\text{rep}} \triangleq \sum_{t=1}^T \mathcal{S}^t_{\text{rep}}$. Under standard concentration of normalized feature means and the Lipschitz continuity of the shared layers h_{ψ_c}, which extract local representations, $\bar{\mathcal{S}}_{\text{rep}}$ accurately tracks the temporal evolution of local feature representations.

Theorem 2 (Error Bound: Cumulative Representation Dynamics)**.** *Assume $\|\mathbf{z}_c^t - \bar{\mathbf{z}}_c^t\|_2 \le \epsilon'_t$, where $\bar{\mathbf{z}}_c^t = \mathbb{E}[h_{\psi_c}(x)/\|h_{\psi_c}(x)\|_2]$, and that h_{ψ_c} is K_h-Lipschitz. Then, the cumulative estimation error is bounded as*

$$\Big|\bar{\mathcal{S}}_{\text{rep}} - \textstyle\sum_{t=1}^T \frac{1}{2}\Big(1 - \frac{\langle \bar{\mathbf{z}}_c^{t-1}, \bar{\mathbf{z}}_c^t\rangle}{\|\bar{\mathbf{z}}_c^{t-1}\|_2\,\|\bar{\mathbf{z}}_c^t\|_2}\Big)\Big| \;\le\; K_h \sum_{t=2}^T (\epsilon'_t + \epsilon'_{t-1}), \tag{16}$$

where ϵ'_t denotes the feature-level approximation error.

Proof. See Appendix D. □

Combined Cumulative Shift Surrogate We define the total cumulative shift surrogate as $\bar{\mathcal{S}}_c = \frac{1}{2}\big(\bar{\mathcal{S}}_{\text{unc}} + \bar{\mathcal{S}}_{\text{rep}}\big)$. Let $\mathcal{V}_c$ denote the corresponding combined true shift measure. Then, by Theorems 1 and 2, and applying the triangle inequality, we obtain the following.

$$\big|\bar{\mathcal{S}}_c - \mathcal{V}_c\big| \;\le\; \tfrac{1}{2}\Big[K_{\cos}\sum_{t=2}^T(\epsilon_t + \epsilon_{t-1}) + K_h\sum_{t=2}^T(\epsilon'_t + \epsilon'_{t-1})\Big]$$

Hence, the combined surrogate $\bar{\mathcal{S}}_c$ closely tracks the true distribution shifts and can be directly used in the subsequent dynamic regret analysis.

5.2. Dynamic Regret Analyses

At each timestep t, client c updates its model parameters $\{\psi_c, \phi_c\}$ and incurs a loss $\mathcal{F}_c^t(\{\psi_c,\phi_c\}) = \mathcal{L}\big(\mathcal{H}(\{\psi_c,\phi_c\}, \mathbf{x}_c^t), \mathbf{y}_c^t\big)$. The *dynamic regret* over T rounds is defined as

$$\text{Reg}_T \triangleq \sum_{t=1}^T \mathcal{F}_c^t(\{\psi_c,\phi_c\}) - \sum_{t=1}^T \min_{\{\psi_c^t,\phi_c^t\}} \mathcal{F}_c^t(\{\psi_c,\phi_c\}),$$

where $\min_{\{\psi_c^t,\phi_c^t\}} \mathcal{F}_c^t$ denotes the instantaneous optimal loss at each round. This formulation quantifies the additional cost incurred by non-stationarity in the local data distribution.

Dynamic Regret Bound Prior analyses of online distribution shift [7, 48, 52] provide an $\mathcal{O}(\sqrt{T})$ *static regret* bound against a fixed comparator. In contrast, in non-stationary environments where client data distributions evolve over time, we establish a *dynamic regret* bound that scales with the cumulative shift proxy $\bar{\mathcal{S}}_c$.

Let $G \triangleq \sup_{(\mathbf{x}_c^t,\mathbf{y}_c^t)} \|\nabla_{\{\psi_c,\phi_c\}} \mathcal{L}(\mathcal{H}(\{\psi_c,\phi_c\},\mathbf{x}_c^t),\mathbf{y}_c^t)\|_2$ be an upper bound on the gradient norm, and $B \triangleq \sup_{(\mathbf{x}_c^t,\mathbf{y}_c^t)} |\mathcal{L}(\mathcal{H}(\{\psi_c,\phi_c\},\mathbf{x}_c^t),\mathbf{y}_c^t)|$ an upper bound on the loss value. Under the assumptions of Lemma 5 (Appendix B), running gradient descent with a constant learning rate η yields the following result.

Theorem 3 (Dynamic Regret Bound)**.** *Let* Reg_T *denote the dynamic regret over* T *rounds. Let* $\sigma > 0$ *be the minimum singular value of the confusion matrix* $\mathbf{M}$*,* Γ *a projection constant, and* $|\mathcal{I}|$ *the number of classes. Then, for learning rate* η*, the expected regret satisfies*

$$\mathbb{E}[\mathrm{Reg}_T] \leq 2\left(\frac{|\mathcal{I}|G^2}{\sigma^2} + B^2\right)\eta T + \frac{\Gamma^2}{\eta} + \frac{4(\Gamma+1)\sqrt{B\bar{\mathcal{S}}_c T}}{\eta}$$
$$\approx \mathcal{O}\left(\eta T + \frac{1}{\eta} + \frac{\sqrt{\bar{\mathcal{S}}_c T}}{\eta}\right). \tag{17}$$

Proof. See Appendix E. □

The three terms ηT, $1/\eta$, and $\sqrt{\bar{\mathcal{S}}_c T}/\eta$ correspond to optimization error, projection cost, and adaptation cost due to distributional shifts, respectively. By tuning the learning rate η with respect to $\bar{\mathcal{S}}_c$, Fed-ADE achieves the min–max optimal dynamic regret rate under non-stationarity.

Min–max Optimality Fed-ADE employs the adaptive learning rate rule (9), parameterized by $[\eta_{\min}, \eta_{\max}]$, where $\mathcal{S}_c^t$ denotes the instantaneous shift signal at round t. Choosing $\eta_{\min}$ and $\eta_{\max}$ such that $\eta_{\min} \leq \eta^* \leq \eta_{\max}$, with $\eta^* = \Theta(T^{-1/3}\bar{\mathcal{S}}_c^{1/3})$ being the theoretically optimal rate, ensures that η_c^t remains within the valid range. This choice satisfies the convergence conditions in Theorem 12 and guarantees that Theorem 3 attains min–max optimality as follows.

$$\mathbb{E}[\mathrm{Reg}_T] = \mathcal{O}\left(\bar{\mathcal{S}}_c^{\frac{1}{3}} T^{\frac{2}{3}}\right). \tag{18}$$

Further description for min-max optimality is provided in Appendix E.2.

Remark 4. The regret rate $\mathcal{O}(\bar{\mathcal{S}}_c^{1/3}T^{2/3})$ achieved by Fed-ADE matches the min–max optimal bound for non-stationary online learning under unsupervised label shift [5]. The adaptive rate bound $[\eta_{\min}, \eta_{\max}]$ is designed to contain the optimal value estimated by $\bar{\mathcal{S}}_c$, aligning theoretical guarantees with practical adaptivity in dynamic federated environments.

6. Simulation Results

In this section, we conduct extensive experiments to validate the effectiveness of the proposed methods. We first briefly introduce the simulation setups and then present numerical results on various datasets. The detailed description of the simulation setups is provided in Appendix G, and results of extra simulations are provided in Appendix H.

6.1. Simulation Settings

Shift Scenarios and Datasets We evaluate Fed-ADE across three prototypical online distribution-shift settings—label shift and covariate shift—using standard image benchmarks (Tiny ImageNet [32], CIFAR-10 [29], CIFAR-100 [29], CIFAR-10-C [19], CIFAR-100-C [19]) and the text benchmark LAMA [50].

- **Label shift** is simulated by gradually changing each client's class prior over time, following one of four temporal schedules (linear (Lin.), sine (Sin.), square (Squ.), Bernoulli (Ber.) on Tiny ImageNet [32], CIFAR-10 [29], and LAMA [50] benchmarks.
- **Covariate shift** is modeled using corruption benchmarks CIFAR10-C [19] and CIFAR100-C [19], where clients observe progressively changing corruption severity levels following one of four temporal schedules (Lin., Sin., Squ., Ber.).

Baselines for Comparing Adaptation Methods We compare the proposed Fed-ADE with the following baselines.

- **Localized Learning:** We evaluate four centralized unsupervised online adaptation methods—FTH [58], ATLAS [2], UNIDA [63], and UDA [18]. These methods perform adaptation locally without any client collaboration, thus lacking the ability to leverage shared knowledge across users.
- **Federated Learning**: We evaluate two federated adaptation methods—Fed-POE [43], and FedCCFA [9] that combine locally fine-tuned models with aggregated global models over time, using a fixed learning rate irrespective of shift magnitude.
- **Fed-ADE (fixed learning rate)**: We evaluate fixed learning rate (FixLR) variants that remove adaptivity from Fed-ADE and use constant stepsizes within the same bounds as Fed-ADE; For image benchmarks, we use 5×10^{-6} for FixLR(Low), 10^{-5} for FixLR(Mid), and 10^{-4} for FixLR(High). For LAMA, we use 10^{-4} for FixLR(Low), 10^{-3} for FixLR(Mid), and 10^{-2} for FixLR(High).
- **Fed-ADE (ours):** Our method performs personalized federated post-adaptation by dynamically adjusting the local learning rate based on estimated distribution-shift magnitude. For all image benchmarks, we set $\eta_{\min} = 5 \times 10^{-6}$ and $\eta_{\max} = 10^{-4}$, while for the text-based LAMA benchmark we use $\eta_{\min} = 10^{-3}$ and $\eta_{\max} = 10^{-2}$.

Evaluation Protocols At each timestep, clients adapt using unlabeled data and are evaluated on the same data with labels, reporting accuracy averaged over clients and timesteps. We assume clients do not know the pre-training distribution and initialize with uniform priors. For simulation, We set the total number of timesteps to 100 and the total number of clients to 100.

6.2. Simulation Results

Performance of Adaptation Methods Table 1 summarizes the results across three prototypical shift scenarios—label and covariate—over diverse datasets. Overall,

Table 1. Performance comparison of different online adaptation methods (average accuracy (%) and average wall time (sec.)).

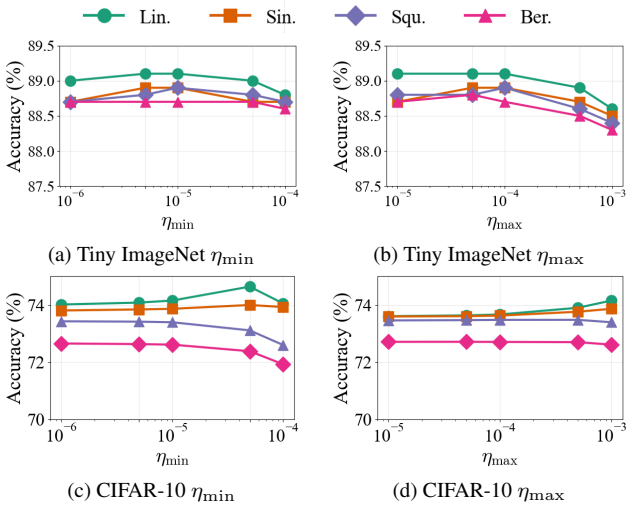
Dataset	Shift	Localized Learning				Federated Learning					
		FTH	ATLAS	UNIDA	UDA	Fed-POE	FedCCFA	FixLR(Low)	FixLR(Mid)	FixLR(High)	**Fed-ADE**
(i) Label Shift Scenarios											
Tiny ImageNet	Lin.	78.2±1.0	76.5±1.3	83.8±0.1	74.5±0.2	87.1±0.2	84.7±0.8	87.5±1.2	88.2±1.1	86.1±0.4	**89.1**±0.1
	Sin.	77.9±0.8	76.8±1.1	83.2±0.5	74.5±0.2	87.5±0.4	84.8±0.7	87.3±1.4	88.0±1.1	87.6±0.6	**88.9**±0.1
	Squ.	77.2±0.8	78.5±1.4	83.3±0.1	74.7±0.3	86.4±0.9	83.0±0.9	87.4±1.2	88.2±0.7	86.4±0.5	**88.9**±0.1
	Ber.	78.2±1.1	77.6±1.1	82.7±0.3	73.8±0.8	86.5±0.7	83.8±0.9	86.5±1.7	87.8±1.2	86.3±0.6	**88.7**±0.1
CIFAR-10	Lin.	31.4±0.8	36.5±4.3	23.0±0.3	33.3±1.4	71.3±3.2	65.8±0.5	70.6±2.0	70.8±2.1	63.8±1.9	**73.8**±0.6
	Sin.	40.3±0.9	43.7±5.1	22.9±0.3	32.0±1.3	71.4±2.6	65.8±0.8	69.4±1.5	70.5±1.6	64.3±2.2	**73.6**±0.5
	Squ.	31.7±0.7	32.3±5.0	23.1±0.1	28.1±1.4	70.6±1.9	65.3±0.3	71.8±2.1	71.6±2.0	70.6±2.5	**72.2**±1.6
	Ber.	30.6±0.9	32.7±5.9	23.0±0.1	28.5±1.5	69.6±1.5	65.4±0.4	68.3±1.7	71.8±1.6	70.0±2.2	**72.9**±2.2
LAMA	Lin.	68.3±1.2	79.5±3.2	31.2±0.8	72.9±2.0	85.4±1.3	95.6±0.1	86.7±1.2	95.2±2.0	24.6±3.3	**95.8**±0.4
	Sin.	74.7±3.1	71.8±5.0	31.1±0.6	70.6±6.8	84.0±1.6	91.6±0.9	88.0±0.8	94.7±2.4	26.9±4.1	**95.8**±0.6
	Squ.	70.5±6.5	79.8±0.9	31.2±0.5	74.4±0.2	84.2±1.0	92.0±0.1	88.6±0.2	95.4±1.2	26.9±5.3	**96.4**±0.6
	Ber.	76.8±0.2	78.0±5.8	31.1±0.5	70.9±6.5	84.1±0.6	91.1±0.5	87.9±0.3	94.3±2.4	20.0±4.7	**95.9**±0.5
(ii) Covariate Shift Scenarios											
CIFAR-10 CIFAR-10-C	Lin.	23.7±0.2	13.9±0.2	43.3±0.6	45.7±0.8	44.5±0.8	43.1±0.5	63.4±0.2	63.9±0.3	40.6±2.1	**64.4**±0.2
	Sin.	22.9±0.2	13.2±0.2	44.8±0.1	43.5±0.3	44.7±1.2	43.3±0.4	63.5±0.3	63.7±0.3	38.1±3.1	**64.8**±0.1
	Squ.	23.8±0.1	14.1±0.4	42.2±0.1	43.3±0.2	48.5±1.2	41.6±0.7	62.7±2.1	64.5±2.1	39.6±2.8	**65.4**±0.2
	Ber.	23.6±0.2	14.2±0.3	42.7±0.2	42.3±0.1	48.7±0.9	42.0±0.5	64.1±2.1	64.4±2.1	40.8±2.2	**65.8**±0.2
CIFAR-100 CIFAR-100-C	Lin.	9.2±0.1	3.5±0.1	34.3±0.1	35.6±0.2	27.3±0.1	24.3±0.3	44.0±0.2	43.4±0.6	41.4±0.6	**45.8**±0.2
	Sin.	7.6 ±0.2	2.9±0.1	34.1±0.1	35.7±0.1	27.5±0.1	24.5±0.5	43.9±0.1	42.1±0.2	39.8±0.3	**46.7**±0.1
	Squ.	8.5±0.1	3.4±0.1	33.8±0.2	35.3±0.1	31.9±0.1	25.1±0.4	44.9±0.3	42.8±0.2	34.3±0.5	**46.5**±0.3
	Ber.	8.6±0.1	3.5±0.1	33.0±0.1	34.7±0.2	31.3±0.1	25.2±0.4	45.3±0.5	43.5±0.2	31.1±0.4	**46.8**±0.1
Average Wall Time		2631.2±71.3	1959.2±22.6	1886.5±3.7	1242.7±7.8	130.7±1.9	210.8±16.6	**109.5**±0.4	**109.5**±0.4	**109.5**±0.4	**109.5**±0.4

Fed-ADE achieves the highest accuracy in all settings while maintaining the lowest wall-clock time. Localized adaptation methods (FTH, ATLAS, UNIDA, UDA) exhibit limited robustness and incur substantially higher computation, whereas federated approaches such as Fed-POE and FedCCFA provide improved stability but still fall short of Fed-ADE in dynamic environments.

Under label distribution shift, Fed-ADE consistently surpasses all baselines across all temporal schedules (Lin., Sin., Squ., Ber.). On Tiny ImageNet and CIFAR-10, Fed-ADE provides steady improvements over FixLR and Fed-POE, with average gains of about 1% and 4%, respectively. A similar pattern is observed on the text-based LAMA benchmark: despite its larger label space and noisier dynamics, Fed-ADE still maintains the highest accuracy across all temporal schedules, improving over the best federated baselines by roughly 2% on average.

In covariate shift scenarios, centralized domain adaptation methods (UDA, UNIDA) struggle in the federated setting, showing notably low robustness across all corruption severities. Fed-ADE consistently surpasses both FixLR and Fed-POE, achieving an average improvement rate of roughly 3% over FixLR and more than 6% over Fed-POE.

In terms of computational cost, Fed-ADE delivers superior performance with exceptional efficiency. Its average wall time (≈ 109 sec) is 17–24$\times$ faster than localized methods and roughly $2\times$ faster than FedCCFA, despite achieving consistently higher accuracy. These results indicate that Fed-ADE provides both accuracy and scalability under diverse online distribution shifts, achieving dynamic robustness without sacrificing computational efficiency—unlike existing approaches that specialize in only one shift type or incur substantial overhead.



(a) Tiny ImageNet $\eta_{\min}$ (b) Tiny ImageNet $\eta_{\max}$

(c) CIFAR-10 $\eta_{\min}$ (d) CIFAR-10 $\eta_{\max}$

Figure 2. Impact of learning rate bounds on Fed-ADE under label shift scenarios on Tiny-ImageNet and CIFAR-10 datasets.

Table 2. Performance with Pre-trained Model Trained on Various Distributions with CIFAR-10.

Distribution	Shift	Fed-ADE	FedCCFA	Fed-POE	UDA	UNIDA	ATLAS	FTH
Gaussian	Lin.	**69.2±1.7**	62.3±1.2	65.7±2.1	20.1±2.0	20.2±1.8	25.3±2.2	22.3±1.7
	Sin.	**68.3±1.6**	63.1±1.5	64.8±1.7	27.8±2.4	27.7±2.2	33.5±2.3	23.9±2.1
	Squ.	**66.4±1.5**	61.9±2.6	61.4±2.2	20.3±2.1	19.8±1.7	23.1±2.0	23.0±1.8
	Ber.	**67.1±1.3**	63.0±2.5	60.9±2.5	18.2±2.3	17.7±1.9	22.5±2.1	21.6±1.9
Exponential Decay	Lin.	**67.4±1.6**	54.1±3.3	61.1±1.8	23.4±3.1	25.4±2.8	26.1±1.5	22.3±1.9
	Sin.	**67.6±1.6**	54.2±2.5	61.4±2.5	23.9±2.9	24.5±2.3	25.6±2.1	21.5±1.7
	Squ.	**66.9±1.5**	52.3±3.2	55.3±2.7	19.2±4.3	21.7±2.0	26.6±2.2	21.4±1.7
	Ber.	**66.7±1.1**	53.0±2.8	57.5±2.3	21.8±2.8	19.3±3.3	25.3±2.7	22.5±2.2

Impact of Learning Rate Selection Figure 2 examines the sensitivity of Fed-ADE to the choice of learning rate bounds under label shift scenarios on CIFAR-10 and Tiny ImageNet. We vary $\eta_{\min}$ while fixing $\eta_{\max} = 10^{-4}$, and vary $\eta_{\max}$ while fixing $\eta_{\min} = 5 \times 10^{-6}$. Across both datasets, Fed-ADE consistently maintains high accuracy within a broad range of hyperparameter choices. While extreme values lead to slight performance degradation, the overall trend shows that our dynamics-driven adaptation is far less sensitive to the exact learning rate bounds compared to fixed-rate baselines. This robustness highlights that Fed-ADE reduces the need for extensive hyperparameter tuning, an essential property in federated and online learning where search budgets are limited.

Impact of Distribution of Pre-training Data In practical federated deployments, the exact distribution of the pre-training data on the server is often unknown to clients. To examine the robustness of Fed-ADE under such realistic conditions, we evaluate post-adaptation performance on CIFAR-10 when the pre-training data follows two non-uniform distributions: *Gaussian* and *Exponential Decay*. As shown in Table 2, Fed-ADE maintains consistently strong performance across all distribution settings and shift types, with only a minor variation compared to the uniform case (cf. Table 1). These results demonstrate that our dynamics-based adaptation effectively generalizes even when the pre-training distribution differs from the assumed prior, validating its robustness under practical deployment scenarios.

Ablation Study on the Similarity Measure in Fed-ADE Table 3 compares cosine similarity with KL divergence [1, 24, 54], Wasserstein distance [37, 57, 67], and Bayesian Check Point Detection (CPD) [14, 17, 66] under four dynamic label-shift schedules on CIFAR-10. As shown in Table 3, our cosine similarity-based Fed-ADE achieves the highest accuracy in all cases. The key reason is that cosine similarity provides a bounded and direction-based measure of drift, making it robust to noisy pseudo-labels and class imbalance that frequently arise under online shifts. In contrast, KL and Wasserstein depend on absolute probability differences and become unstable when class probabilities are near zero, while Bayesian CPD often overreacts to transient fluctuations. Thus, cosine similarity-based Fed-ADE offers the most stable shift signal, enabling Fed-ADE to adjust learning rates reliably across clients. The detailed description of this ablation study is provided in Appendix H.3.

Table 3. Ablation study on the choice of distribution-shift similarity measure in Fed-ADE under CIFAR-10 label-shift scenarios.

Measure	Lin.	Sin.	Squ.	Ber.
Fed-ADE (Cosine similarity)	**73.8 ± 0.6**	**73.6 ± 0.5**	**72.2± 1.6**	**72.9 ± 2.2**
KL divergence	63.2 ± 3.8	62.4 ± 2.3	61.1 ± 3.4	61.4 ± 3.6
Wasserstein	70.8 ± 0.6	69.9 ± 1.1	68.8 ± 1.3	66.5 ± 2.1
Bayesian CPD	71.5 ± 0.5	70.8 ± 0.8	69.7 ± 2.4	69.0 ± 1.1

Table 4. Ablation of Fed-ADE estimators on CIFAR-10

S_{unc}	S_{rep}	Variant	Label Shift (%)	Covariate Shift (%)
✓	✓	**Fed-ADE (full)**	73.8 ± 0.6	64.4 ± 0.2
✗	✓	w/o S_{unc}	71.3 ± 0.3	64.2 ± 0.1
✓	✗	w/o S_{rep}	73.1 ± 0.4	64.0 ± 0.3
✗	✗	Fixed LR	70.8 ± 2.1	63.9 ± 0.3

Table 4 shows that both w/o S_{unc} and w/o S_{rep} underperform the full Fed-ADE across label and covariate shifts, while a fixed learning rate performs worst. This indicates that S_{unc} and S_{rep} capture complementary distribution dynamics and are both necessary for effective learning rate adaptation.

Ablation Study on Estimators in Fed-ADE Table 4 shows that removing either estimator degrades performance under both label and covariate shifts. In particular, w/o S_{unc} leads to a noticeable drop under label shift, while w/o S_{rep} slightly reduces performance under covariate shift. The fixed learning-rate baseline performs worst and exhibits higher variance. These results indicate that S_{unc} and S_{rep} capture complementary aspects of distribution dynamics, and that combining both signals yields more stable and effective learning-rate adaptation.

7. Conclusions

We presented Fed-ADE, a novel framework for unsupervised federated post-adaptation under evolving distribution shifts. Our approach combines lightweight estimators for online distribution shifts with a theoretically grounded adaptive learning rate and a personalized federated update scheme, providing both provable dynamic regret and convergence guarantees and high practical efficiency. Beyond its methodological novelty, Fed-ADE is validated through extensive experiments on various benchmarks, covering major shift types where it consistently outperforms strong baselines. This breadth of validation highlights the robustness, versatility, and real-world readiness of our method, positioning Fed-ADE as a principled and practical foundation for advancing adaptive, shift-aware federated learning in dynamic and heterogeneous environments.

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